Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/20/2019
PI Name:	Narayanan, Ranga Ph.D.		
Project Title:	A Novel Way to Measure Interfacial Tension Using the Electrostatic Levitation Furnace (ELF)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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	A novel method using an Electrostatic Levitator (EL) is proposed to determine the interfacial tension (IFT) of liquid/gas systems. This key thermophysical property is of interest for materials processing and in welding operations in space and on ground. The novel method for measurement of IFT proposed involves the use of electrostatically forced Faraday instability. In this method, an oscillatory electrostatic forcing is applied to a liquid system, resulting in sudden pattern formation at the interface, i.e., an instability is caused. The "instability" is a result of resonance between the imposed forcing frequency and the natural frequency of the system. The critical amplitude and mode shapes are predictable from linear stability theory and agree superbly with experiments. The measurement of IFT directly aligns with the goals of the Thermophysical Property Measurement emphasis of the current NRA (NASA Research Announcement). Faraday experiments by this team using mechanical forcing have shown excellent agreement between theory and		

Task Description:	experimental results. However, in the case of mechanical forcing, the force of gravity overwhelms the effects of IFT on the system stability and therefore it is almost impossible to determine the IFT with accuracy. Electrostatic forcing counters gravity, allowing the effects of IFT to play a significant role in the ultimate determination of the system stability. Accordingly, experimentally mapped stability curves can be fit accurately with theoretical curves, allowing for the determination of the IFT with great accuracy. The proposing team has successfully used electrostatic forcing in ground equipment to show that Faraday waves can be obtained as per theoretical predictions. The current state of the art uses an open container geometry to hold the liquid. However, a levitated sphere may also be used, and our proposal necessarily includes this method. We therefore propose a project with two NEW aspects. In the first stage of the research, we propose to use ground-based facilities at JAXA (Japan Aerospace Exploration Agency) and NASA Marshall Space Flight Center (MSFC) to accurately obtain the IFT and surface tension for liquids using electrostatic Faraday forcing in open containers as well as in levitated drops. In the second stage of the research, we will perform flight experiments using the Electrostatic Levitation Furnace (ELF) in the International Space Station (ISS) to obtain the threshold for instability and determine the surface tension of these systems. In summary, the research allows an alternate new method, viz., Faraday instability, for the determination of interfacial tension. Using electrostatic forcing instability will permit an independent confirmation of measurements obtained by the single-pulse oscillating-drop technique where decay rates are measured and related to interfacial tension and viscosity. Ground-based research using the electrostatic levitator available at the NASA MSFC and JAXA, and space experiments on ELF in the ISS will provide accurate data. In addition to being an alternative method for IFT measurement, the proposed technique also has versatility as a potential means in measuring interfacial tension for high viscosity materials and two-liquid interfacial tension on ground as well as in space. The proposing team has recently collaborated successfully with JAXA and has conducted experiments to study the feasibility of electrostatic Faraday instability. These results form the basis for the current proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Accurate interfacial tension values are crucial in processes of importance to both industry and international space missions, including: a) Additive manufacturing of metals, where IFT knowledge is pivotal to avoid shrinkage and wrinkling. Human Exploration: TA7.2.3.5: Welding, TA12.1.2: Computationally-Designed Materials, TA12.4.2.5: Additive Manufacturing. b) Semiconductor crystal growth, where the IFT between a liquid melt and liquid encapsulant plays a large role in determining conditions where convection is present. c) Space enabling operations. One example is joining operations such as soldering, where the IFT plays an important role in the structure and stability of the weld. Accurate values for the IFT will enhance the safety and effectiveness of this type of operation in space where the microstructures formed are different than those on Earth where gravity is present. d) Enhanced oil recovery, where the IFT between injection fluid and crude oil plays an important role in the recovery rate of the oil and the prevention of viscous fingering. e) The accurate measurement of the IFT between liquids through the use of a different operational mode with minimal experimental modification can validate measurements made through the current techniques of levitation and pulse disturbance of spherical fluid masses.
Task Progress:	A novel measurement method for the measurement of surface tension of high temperature liquid metals has been developed with a proof-of-concept. The method involves Faraday forcing. This is a means wherein a fluid is mechanically or electrostatically oscillated over a frequency range that includes the natural frequency. When the parametric frequency equals the natural frequency, the system undergoes an instability that is manifested by a sudden change of its configuration. The natural frequency is related to the surface tension and so the onset of the instability is used to determine the surface tension. This technique was demonstrated for a liquid levitated drop. It has also been demonstrated for a liquid bilayer system. In the case of the latter, the instability is manifested by the sudden generation of interfacial waves. In the case of liquid bilayers the instability threshold is used to determine the interfacial tension. The knowledge of these thermophysical properties is extremely useful to predictive additive manufacturing.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Brosius N, Ward K, Matsumoto S, SanSoucie M, Narayanan R. "Faraday forcing of high-temperature levitated liquid metal drops for the measurement of surface tension." NPJ Microgravity. 2018 May 14;4:10. eCollection 2018. https://doi.org/10.1038/s41526-018-0044-1 ; PubMed PMID: 29796410; PubMed Central PMCID: PMC5951803 , May-2018
Articles in Peer-reviewed Journals	Ward KL, Matsumoto S, Narayanan R. "The electrostatically forced Faraday instability: theory and experiments." Journal of Fluid Mechanics. 2019 Mar 10;862:696-731. https://doi.org/10.1017/jfm.2018.940 , Mar-2019
Articles in Peer-reviewed Journals	Pillai DS, Narayanan R. "Nonlinear dynamics of electrostatic Faraday instability in thin films." Journal of Fluid Mechanics. 2018 Nov 25;855: R4. https://doi.org/10.1017/jfm.2018.682 , Nov-2018
Dissertations and Theses	Ward KL. "Faraday Instability in Mechanically and Electrostatically Forced Systems." Dissertation, University of Florida, May 2018. , May-2018